Ramatoulaye Bah

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Education

Macalester College St. Paul, MN

B.A. in Computer Science and Linguistics, Minor in Spanish, ***GPA: 3.9/4.00*** *Expected May 2027*

Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Software Design and Development, Computer Systems, Artificial Intelligence, Natural Language Processing, CodePath iOS and Web Development
Honors and Awards: Dean's List, Davis Scholar, Kofi Annan Scholar, Inclusive Coding Festival 2026 First Place Award
Organizations: Macalester Women's Soccer, ColorStack, Rewriting The Code, Girls Who Code, GameDevs at Mac

Work Experience

Software Engineer May 2026 – Aug. 2026

Demystify.org *Twin Cities, MN*

- Founded a tax-literacy platform with **Next.js** that translates uploaded documents into simplified explanations for Somali speakers, validated through **5 user-testing sessions** and feedback from **20+** community members within 10 weeks.
- Architected a single-call **Claude** vision pipeline that classifies and extracts **181+** fields across **23** IRS form types, which allows multimodal uploads with no file persistence for user privacy.
- Developed a bilingual Somali/English chatbot with scope controls, cutting API costs by ∼**40%** through prompt caching and a sliding 20-message window.

Software Engineer Intern May 2025 – Dec. 2025

Gobo Show, Inc. *Minneapolis, MN*

- Reduced Events page load times by **30%** for an event-ticketing platform serving independent artists and venues by migrating the application from **Vue 2** to **Next.js** with server-side rendering, **TypeScript**, and **Tailwind CSS**.
- Optimized artist onboarding flow with **React** and **Firebase Firestore** by streamlining account creation, anonymous authentication, and invitation-based signup, increasing user conversion by ∼**25%**.

Software Engineering Fellow July 2024 – Sept. 2024

Headstarter AI *Remote*

- Shipped 5 production-ready full-stack **AI applications** in 5 weeks using **React**, **Next.js**, and Firebase with **OpenAI**, **Pinecone**, and **Stripe** integrations under **CI/CD** practices, reaching **200+** users.
- Led 4 engineering fellows through **MVC-pattern** architecture decisions from design to deployment, reducing PR cycle time by ∼**15%**, through structured code reviews and clear ownership boundaries.

Projects

Lock The Code | https://lockthecode.net/ | *Next.js, TypeScript, FastAPI, PostgreSQL, Python, Anthropic API*

- Engineered a full-stack interview-prep SaaS in **Next.js** and **TypeScript**, with a **FastAPI** backend, using **PostgreSQL** to power spaced-repetition scheduling, onboarding **70+** users and converting 1 paying customer within 2 weeks of launch.
- Built a streaming AI tutor and mock-interviewer with the **Anthropic API** and integrated **Clerk** auth with **Stripe** billing, handling idempotent webhooks to keep subscription state consistent across payment retries.
- Validated **REST API** reliability with a **15-test pytest** integration suite achieving **100%** endpoint coverage on highest-risk workflows, automated via **GitHub Actions CI** and **Docker Compose** for reproducible local development.

LingoHub | https://lingo-hub.vercel.app/ | *React, Next.js, JavaScript, Supabase, PostgreSQL, Jest*

- Migrated a full-stack forum for language learners from **Firebase** to **Supabase** to support relational **SQL** queries, enabling real-time posts, comments, upvotes, and saved collections for **35+** active users.
- Established a testing foundation with **Jest** and **React Testing Library**, writing tests across post rendering, navigation, and auth state flows to catch regressions early in the development cycle.

Social Network Visualizer | Link | *Java, PageRank, Graph Theory, Data Structures*

- Designed a weighted **Java** graph using HashMap adjacency lists and CSV data to model relationships from shared attributes.
- Implemented Google's **PageRank algorithm** to rank influence and generate a top-5 leaderboard with an interactive graph.

Technical Skills

Languages: Python, Java, C, TypeScript, JavaScript, SQL, HTML, CSS, Swift
Frameworks: React, Next.js, Node.js, FastAPI, Tailwind CSS, Bootstrap, Material-UI
Cloud & Tools: Git, GitHub Actions, Docker, Vercel, Railway, GCP, Cloudflare, Clerk, PostHog, Figma, Vite, Linux
AI/ML & APIs: PyTorch, NumPy, Hugging Face, Claude Code, Anthropic API, OpenAI API, Pinecone, REST APIs, NetLogo